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THE FLOOR – minimum sensitivity under cyclic symmetry

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First, a simplification that makes this tractable and that is worth stating on its own:

If f is cyclically invariant and does not depend on variable i , then by rotating it depends on NO variable, so f is constant.

So for these functions "non-degenerate" and "non-constant" are the same condition. And sensitivity is constant across each necklace, so we only evaluate one point per necklace instead of all 2^n points. That is the speedup that makes $n = 7$ exhaustive at all.

[1] EXHAUSTIVE – every cyclically invariant function

n	necklaces	functions	min s	#minimisers	status
3	4	16	2	2	EXACT
4	6	64	2	4	EXACT
5	8	256	3	18	EXACT
6	14	16,384	3	36	EXACT
7	20	1,048,576	3	28	EXACT

[2] SEARCH – beyond exhaustive reach

$n \geq 8$ has 2^{36} or more functions. We anneal instead, which gives an UPPER BOUND on the true minimum. Many restarts, best kept.

n	necklaces	functions	best s found	status
8	36	2^{36}	4	BOUND ($\leq$)
9	60	2^{60}	5	BOUND ($\leq$)
10	108	2^{108}	6	BOUND ($\leq$)
11	188	2^{188}	8	BOUND ($\leq$)
12	352	2^{352}	9	BOUND ($\leq$)

[3] THE TABLE, AND WHAT IT TRACKS

n	min s	kind	$n^{(1/3)}$	$\sqrt{n}$	divisors of n
3	2	EXACT	1.44	1.73	[1, 3]
4	2	EXACT	1.59	2.00	[1, 2, 4]

5	3	EXACT	1.71	2.24	[1, 5]
6	3	EXACT	1.82	2.45	[1, 2, 3, 6]
7	3	EXACT	1.91	2.65	[1, 7]
8	4	<=	2.00	2.83	[1, 2, 4, 8]
9	5	<=	2.08	3.00	[1, 3, 9]
10	6	<=	2.15	3.16	[1, 2, 5, 10]
11	8	<=	2.22	3.32	[1, 11]
12	9	<=	2.29	3.46	[1, 2, 3, 4, 6, 12]

[4] THE HYPOTHESIS THE TABLE SUGGESTS

Look at the divisor column against the floor. Cyclic symmetry gives a function freedom exactly where the shift has SHORT orbits, and short orbits exist precisely when n is composite. So the prediction is:

composite n should permit a LOWER floor than nearby primes.

Testing that directly on our own numbers:

prime n	floor	composite n	floor
3	2	4	2
5	3	6	3
7	3	8	4
11	8	9	5
		10	6
		12	9

mean floor, prime n : 4.00

mean floor, composite n : 4.83

[5] WHAT THE MINIMISERS ACTUALLY LOOK LIKE ($n = 6$, exact)

36 functions achieve $s = 3$. First four, by necklace:

['000000', '000001', '000011', '000101', '000111']

Hamming weights of chosen necklaces: [0, 1, 2, 2, 3]

['000000', '000001', '000011', '000101', '001001', '001011']

Hamming weights of chosen necklaces: [0, 1, 2, 2, 2, 3]

['000000', '000001', '000011', '000101', '000111', '001001', '001011']

Hamming weights of chosen necklaces: [0, 1, 2, 2, 2, 3, 3]

['000000', '000001', '000011', '000101', '001001', '001101']

Hamming weights of chosen necklaces: [0, 1, 2, 2, 2, 3]

They are weight-threshold-like: the chosen necklaces cluster at one end of the weight scale. That is the same shape as a MAJORITY-style function, which is the classic low-sensitivity object. The cyclic constraint is bending majority, not replacing it.

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VERDICT

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EXACT values obtained : $n = 3..7 \rightarrow [2, 2, 3, 3, 3]$
BOUNDS obtained : $n = 8..12 \rightarrow [4, 5, 6, 8, 9]$

Is this a novel discovery? Honest answer: NO, and here is the specific reason. The floor we measure is growing, slowly, and the literature already contains a construction (Chakraborty, 2005) achieving sensitivity on the order of $n^{1/3}$ for cyclically invariant functions at large n . Our $n \leq 14$ is far too small to distinguish $n^{1/3}$ from $\log n$ from anything else -- look at how flat the table is.

What we DO have that is real:

- * an exact table for $n \leq 7$ that anyone can check in seconds
- * a clean lemma (non-degenerate = non-constant here) that made it fast
- * a testable structural hypothesis about composite vs prime n
- * the minimisers identified as bent majority functions

That is a proper small experiment. It is not a theorem, and I am not going to dress it up as one.

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